

Laxmi Charitable Trust's
Sheth L.U.J College of Arts s Sir M.V. College of Science and
Commerce
Department of Information Technology (B.Sc. I.T Semester V)
MERN STACK
Practical – IV

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Class: TYIT	Batch: 1
Date of Assignment: 25-07-2026	Date/Time of Submission: 25-07-2026

4.React Components, Props, and State

a. Create multiple components and render them on a single page.

Code :

Header.jsx

```
function Header() {  
  return (  
    <div>  
      <h1>Placement Management System</h1>  
      <hr />  
    </div>  
  );  
}  
export default Header;
```

Student.jsx

```
function Student() {  
  return (  
    <div>  
      <h2>Student Details</h2>  
      <p>Name : sachin ramesh mehta</p>  
      <p>Roll No : 101</p>  
      <p>Course : B.Sc. IT</p>  
      <p>Semester : VI</p>  
      <hr />  
    </div>  
  );  
}  
export default Student;
```

Eligibility.jsx

```
function Eligibility() {  
  const percentage = 75;  
  
  return (  
    <div>  
      <h2>Eligibility</h2>
```

```
    <p>Percentage : {percentage}%</p>
    <p>Status : {percentage >= 60 ? "Eligible" : "Not Eligible"}</p> <hr />
  </div>
); }
export default Eligibility;
```

Placementstatus.jsx

```
function Placementstatus() {
  const placed = true;
  return (
    <div>
      <h2>Placement Status</h2>
      <p>{placed ? "Placed" : "Not Placed"}</p>
      <hr />
    </div>
  );
}
export default Placementstatus;
```

Companyoffer.jsx

```
function Companyoffer() {
  return (
    <div>
      <h2>Company Offer</h2>
      <p>Company : Infosys</p>
      <p>Package : ₹6 LPA</p>
      <p>Location : Pune</p>
      <hr />
    </div>
  );
}
export default Companyoffer;
```

Skillprogress.jsx

```
function Skillprogress() {
  const react = 90;
  const java = 85;
  const python = 95;
  return (
    <div>
      <h2>Skill Progress</h2>
      <p>React : {react}%</p>
      <p>Java : {java}%</p>
      <p>Python : {python}%</p>
      <hr />
    </div>
  );
}
```

```
);  
}  
export default Skillprogress;
```

Footer.jsx

```
function Footer() {  
  return (  
    <div>  
      <p>© 2026 Sheth L.U.J. & Sir M.V. College</p>  
    </div>  
  ); }  
export default Footer;
```

App.jsx

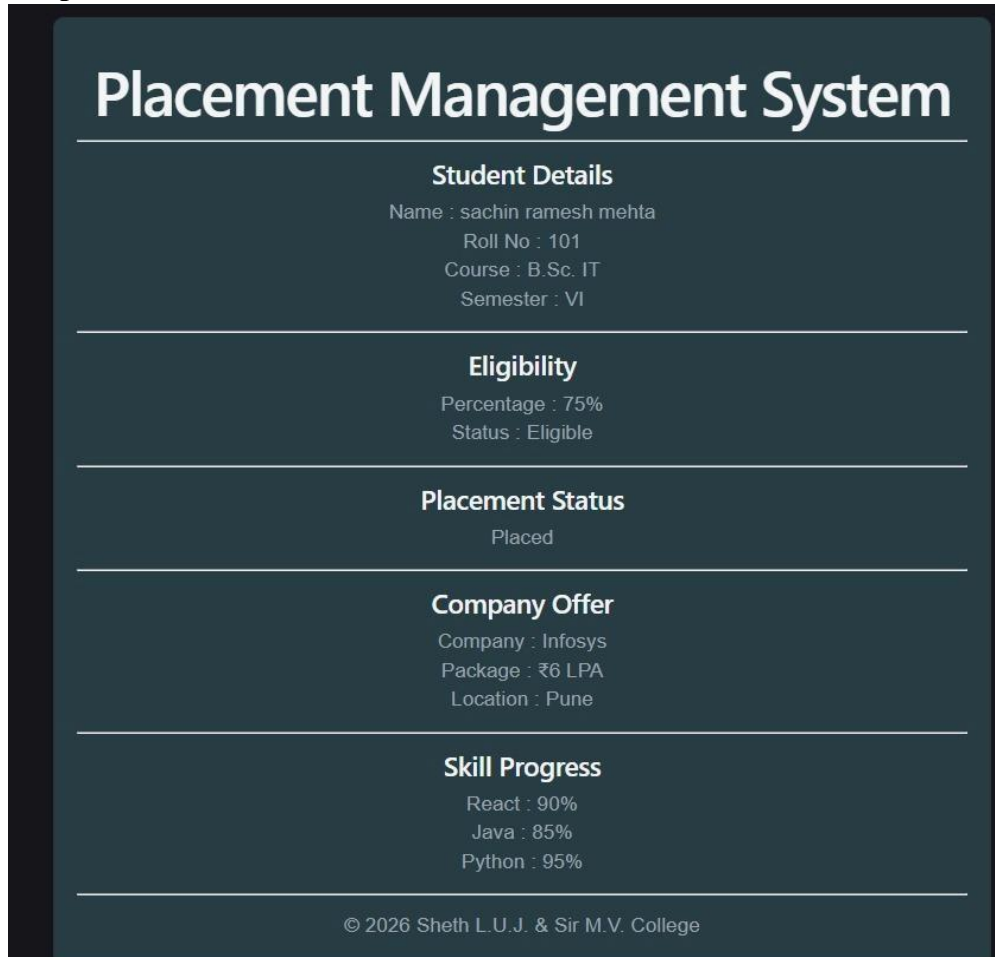
```
import Header from "../components/Header";  
import Student from "../components/Student";  
import Eligibility from "../components/Eligibility";  
import Placementstatus from "../components/Placementstatus";  
import Companyoffer from "../components/Companyoffer";  
import Skillprogress from "../components/Skillprogress";  
import Footer from "../components/Footer";  
import "../App.css";  
function App() {  
  return (  
    <div className="container">  
      <Header />  
      <Student />  
      <Eligibility />  
      <Placementstatus />  
      <Companyoffer />  
      <Skillprogress />  
      <Footer />  
    </div>  
  );  
}  
export default App;
```

App.css

```
.container { width: 600px; margin:  
  30px auto; padding: 20px; border:  
  2px solid #333; border-radius:  
  10px; background-color: #f5f5f5;  
  font-family: Arial, sans-serif;  
}  
  
h1, h2 {
```

```
text-align: center; }  
p { font-size: 18px;  
}  
hr { margin: 15px 0;  
}
```

Output :



b. Pass data between components using props.

Code :

App.jsx

```
import Header from "./components/Header";  
import Footer from "./components/Footer";  
import HotelCard from "./components/HotelCard";  
import "./App.css";
```

```
function App() {  
  return (  
    <div className="container">  
      <Header />  
      <HotelCard  
        restaurant = "Domino's Pizza"  
        food = "Pizza, Pasta, Beverages"  
        price = {1599}  
      />  
    </div>  
  );  
}
```

```
        rating = {4.5}
        discount = {20}/>
    <HotelCard
        resturant = "McDonald's"
        food = "Burgers, Fries, Beverages"
        price = {1699}
        rating = {4.5}
        discount = {20}/>
    <HotelCard
        resturant = "Pizza Hut"
        food = "Pizza, Pasta, Beverages"
        price = {599}
        rating = {4.5}
        discount = {20}/>
    <HotelCard
        resturant = "Chicken Hut"
        food = "chicken Pizza, chicken hande Pasta, Beverages"
        price = {1099}
        rating = {4.5}
        discount = {20}/>
    <Footer />
</div>
);
}
export default App;
```

Header.jsx

```
function Header() {
    return (
        <div>
            <h1>Online Food Delivery Dashboard</h1>
            <h3>Sachin Ramesh Mehta</h3>
            <hr />
        </div>
    );
}
export default Header;
```

HotelCard.jsx

```
function HotelCard(props) {
    const discount = (props.price * props.discount) / 100;
    const finalPrice = props.price - discount;

    return(
        <div>
            <h2>{props.restaurant}</h2>
            <p>Food Item : {props.food}</p>
            <p>Original Price : ₹{props.price}</p>
```

```
        <p>Discount : {props.discount}%</p>
        <p>Final Price : ₹{finalPrice}</p>
        <p>Rating : {props.rating}</p>
      </div>
    );
  }
  export default HotelCard;
```

Footer.jsx

```
function Footer() {
  return (
    <div>
      <p>© 2026 Online Food Delivery App by Sachin Mehta</p>
    </div>
  );
}
export default Footer;
```

Output:

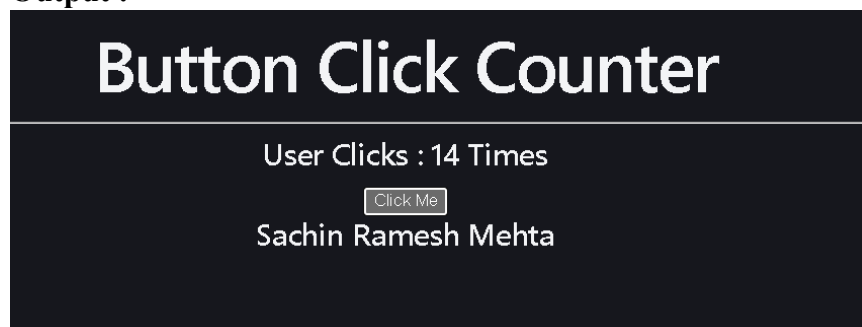


c. Use the useState hook to manage and update data.

Code :

```
import {useState} from "react";
function App(){
  const [count, setCount] = useState(0);
```

```
return(  
  <div>  
    <h1>Button Click Counter</h1>  
    <hr />  
    <h2>User Clicks : {count} Times</h2>  
    <button onClick={() => setCount(count + 1)}>Click Me</button>  
    <h2>Sachin Ramesh Mehta</h2>  
  </div>  
);  
}  
export default App;
```

Output :

d. Create a form component to accept user input and display it dynamically.

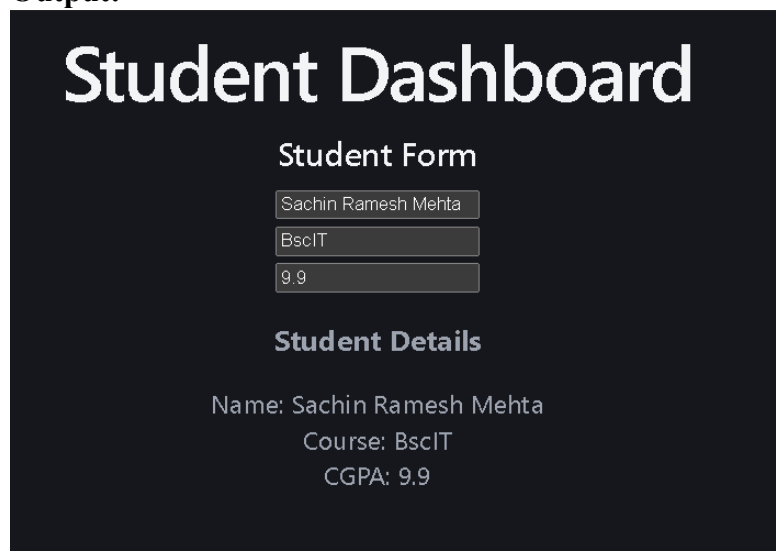
Code:**Student.jsx:**

```
import { useState } from "react";  
function StudentForm() {  
  const [name, setName] = useState("");  
  const [course, setCourse] = useState("");  
  const [CGPA, setCGPA] = useState("");  
  return (  
    <div className="form-container">  
      <h2>Student Form</h2>  
      <input  
        type="text"  
        placeholder="Enter Name"  
        value={name}  
        onChange={(e) => setName(e.target.value)}  
      />  
      <br />  
      <input  
        type="text"  
        placeholder="Enter Course"  
        value={course}  
        onChange={(e) => setCourse(e.target.value)}  
      />  
      <br />  
    </div>  
  );  
}
```

```
<input
  type="text"
  placeholder="Enter CGPA"
  value={CGPA}
  onChange={(e) => setCGPA(e.target.value)}
/>
<br />
<div className="output">
  <h3>Student Details</h3>
  <p>Name: {name}</p>
  <p>Course: {course}</p>
  <p>CGPA: {CGPA}</p>
</div>
</div>
);
}
export default StudentForm;
```

App.jsx:

```
import Student from "../components/Student";
function App() {
  return (
    <div>
      <h1>Student Dashboard</h1>
      <Student/>
    </div>
  );
}
export default App;
```

Output:

The screenshot shows a dark-themed web application titled "Student Dashboard". It features a "Student Form" section with three input fields containing the values "Sachin Ramesh Mehta", "BscIT", and "9.9". Below this is a "Student Details" section displaying the same information: "Name: Sachin Ramesh Mehta", "Course: BscIT", and "CGPA: 9.9".

e. Display a list of items stored in the state using `map()`.
Code:

App.jsx:

```
import CompanyList from "../components/CompanyList";
function App() {
  return (
    <div>
      <h1>Dashboard</h1>
      <CompanyList />
    </div>
  );
}
export default App;
```

CompanyList.jsx:

```
import { useState } from "react";
function CompanyList() {
  const [companies] = useState([
    "TCS",
    "Infosys",
    "Accenture",
    "Capgemini"
  ]);
  return (
    <div>
      <h2>Placement Companies</h2>
      {companies.map((company, index) => (
        <p key={index}>
          {index + 1}. {company}
        </p>
      ))}
    </div>
  );
}
export default CompanyList;
```

Output:**OR**

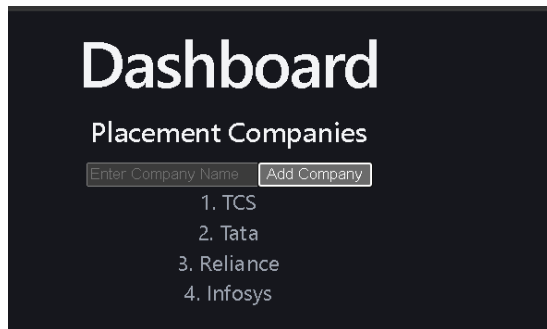
Code:**App.jsx:**

```
import CompanyList from "../components/CompanyList";
function App() {
  return (
    <div>
      <h1>Dashboard</h1>
      <CompanyList />
    </div>
  );
}
export default App;
```

CompanyList.jsx:

```
import { useState } from "react";
function CompanyList() {
  const [companies, setCompanies] = useState([]);
  const [newCompany, setNewCompany] = useState("");
  const addCompany = () => {
    if (newCompany.trim() !== "") {
      setCompanies([...companies, newCompany]);
      setNewCompany("");
    }
  };
  return (
    <div>
      <h2>Placement Companies</h2>
      <input
        type="text"
        placeholder="Enter Company Name"
        value={newCompany}
        onChange={(e) => setNewCompany(e.target.value)}
      />
      <button onClick={addCompany}>Add Company</button>
      {companies.map((company, index) => (
        <p key={index}>
          {index + 1}. {company}
        </p>
      ))}
    </div>
  );
}
export default CompanyList;
```

Output:



f. Build a small interactive component (e.g., counter or to-do list).

Code:

```
import {useState} from 'react';
function App() {
  const [count, setCount] = useState(0);
  return (
    <div className="App">
      <h1>React Counter App</h1>
      <h2>User Clicks : {count} times</h2>
      <button onClick={() => setCount(count + 1)}>Increment</button>
      <br/>
      <button onClick={() => setCount(count > 0 ? count - 1 : 0)}>Decrement</button>
      <br/>
      <button onClick={() => setCount(0)}>Reset</button>
      <hr/>
      <h3>Sachin Ramesh Mehta</h3>
    </div>
  );
}
export default App;
```

Output:

